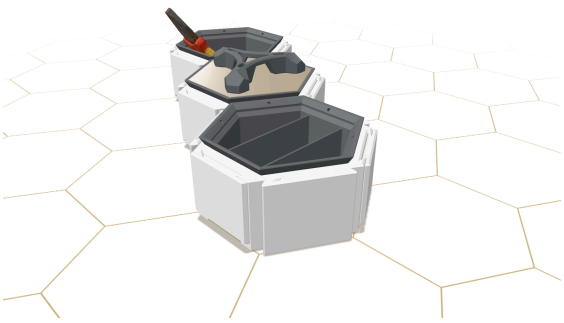
▸ META GLOBAL AI DEVELOPER HACKATHON · STARTUP TRACK · PUSH THE FRONTIER

A Muse Spark 1.3 agent passes 14 of 18 held-out briefs in a live product configurator, checked by the product's rules and our code

hex-agent. Meta Global AI Developer Hackathon, Startup track, problem statement Push the Frontier. One Thousand Drones. Accurate as of 2026-11-03.

hex-agent takes a sentence and returns a build in One Thousand Drones' live Hex Cluster configurator: a share link, a parts list and a print pack made from the catalog's printable parts, a build that passes the configurator's placement and collision rules, and a statement of which requirements were met. On the frozen agent (git tag `eval-final`), Muse Spark 1.3 through the Meta Model API passed 14 of 18 held-out briefs (77.8%) on their one pre-committed run (Wilson 95% interval 54.8 to 91.0%): 11 builds, and 3 declines of briefs written to be refused. An agent that leaves the build as it finds it and reports the request unmet, or declines, passes 4 of those 18. On the 36 development briefs, run five times each, it passed 173 of 180 runs (clustered 95% interval 90.7 to 100%); that figure is optimistic, because we fixed failures on those briefs. A development build cost about $0.057 of Muse Spark 1.3 tokens, an estimate at list price (no bill). A second Meta model, abstention, Meta models as vision judges and the earlier open-weights model are in Demo and results, each reported apart and never pooled. Data files are labelled D1 to D8 and, for results added after the paper's first review gate, A1 to A7, listed in the last section; short hashes name configurator releases and agent commits.



▸ BRIEF

A tool caddy with a carry handle and divided bins for screws and bits.

▸ THE MODEL'S REPORT

Tool caddy form factor (portable, multi-bin base)	MET
Carry handle	MET
Divided bins for screws and bits	MET

▸ CHECKED IN CODE

finished	PASS
claims match checks	PASS
claims in scope	PASS
placement rules	PASS
collision check	PASS
print pack	PASS
share link round trip	PASS
bom handle, min 1: saw 1	PASS
bom div, min 1: saw 2	PASS

PLATE One run from Fireworks F1 (full report, Table A4): Muse Spark Open, effort low, agent 1e98dbf, contract 87827f4d, 16 model calls. Image: the configurator's own still() of the build reloaded from the run's share link. Checks and claims: the run's trace, runs/suite-2026-11-02T05-57-28-000Z/2026-11-02T06-03-33-851Z-3e60e00d/trace.json.

01 WHAT WAS BUILT DURING THE HACKATHON

Pre-existing, One Thousand Drones' product, not judged: the generation pipeline behind Hex Cluster (FreeCAD part authoring, Blender prop generation, glyph and technical-drawing generation), Hex Cluster, its configurator, placement rules, collision check, the public `window.hexAPI`, print packs, printable parts and plant and prop models, plus the configurator releases One Thousand Drones shipped during the event (each identified by its API-contract hash in Table 1 and the report), some prompted by the agent's runs; they are product changes, excluded from the claimed work, and every result here ran on one of them. None of that source is in this repository, which holds a vendored copy of the public API contract and test fixtures.

Built during the hackathon (2026-10-27 to 2026-11-04), in this repository from its first commit (fb78560, 2026-10-27): the agent loop, browser bridge, checked tool layer, Meta model adapters, verification, honesty check, abstention, grader and rescoring, spend ledgers, evaluation suites, baselines, the replay tool and a vision judge on Meta models. Nothing was trained. Every agent result here is a Meta model. Claude (Anthropic) served as the agent's model in 1,325 debugging runs of the loop and as the reviewer agents whose verdicts are the grader audit (D7) and the demo review (D8). It was also the first advisory vision critic; Muse Spark 1.3 and Muse Wonder 1.0 now fill that role (A3), and the Claude critic rows are kept as history. HACKATHON_DELTA.md lists the pre-event product versions.

02 IMPACT

Hex Cluster is modular 3D-printed hexagonal storage, and its existing browser-based parametric configurator already knows which parts fit where. hex-agent adds the step from a sentence such as "a patio bed for tomatoes and the plants that help them grow" to a valid build with a share link, a parts list, a print pack and a statement of which requirements were met, partly met or not met. That is the Push the Frontier claim: a manufacturing tool that took a person's clicks now takes their intent and returns a print pack the configurator accepts; nothing has been printed yet, and its overclaims are measured in Table 1. A development build cost about $0.057 of Muse Spark 1.3 tokens at list price (Table 1). It is built for makers who know what they want but not the part catalog. hex-agent extends the configurator with natural-language autonomous design, evaluated experimentally as reported here; customer-facing deployment in the configurator is planned, and commercial adoption and usage metrics have not yet been established. Hex Cluster is itself one output of One Thousand Drones' existing generation pipeline (pre-existing, not judged): its parts are authored parametrically in FreeCAD by scripted pipelines, its plant and prop models are generated in Blender, and its part glyphs and technical drawings are generated from the same CAD. The pipeline is designed to go further, and these steps are designed or partly built, not demonstrated here: redefining the system's geometry (a cube or pyramid cell in place of the hexagon) with its attachment logic and tolerances, and adding new interface types with their CAD, drawings, reports, assembly guides and print orientation, registered automatically in the configurator's API, in this agent's tools and in the configurator. hex-agent takes its operations and value lists from the configurator's published API contract (a vendored copy, refreshed per release) and its parts, sockets and rule verdicts from the live configurator, so a new product family would reach the agent partly through the contract; the tool layer still hard-codes hex axial coordinates and Hex Cluster part hints (src/tools.mjs), which would have to be generated as well. That is the larger frontier, and it is untested. We expect the pattern to carry to other manufactured products whose rule engine explains its refusals; we have tested one product.

03 USE OF META AI TOOLING

Three Meta models do the design work, measured apart, and two of them now also judge it from images. Muse Spark 1.3 and Muse Wonder 1.0 on the Meta Model API, and the open-weights Muse Spark Open on partner hosts, each read the build, ask where parts fit and place them through 17 tools on the frozen agent (fewer on the earliest commits) over up to 30 turns, reading each refusal and the alternatives the configurator offers. Beyond tool calling, three capabilities are measured. Self-report: every requirement marked met, partial or unmet, with all-met claims checked.

Abstention: declining a brief that would carry a child or hold fuel, or building only its safe part. Its prompt block has an effect beyond the hazards it names: on 12 unsafe briefs on hazards the block does not name, Muse Spark 1.3 passed 27 of 36 runs with it on and 12 of 32 with it off (per-brief means 75% and 44.4%; +30.6 points paired, 95% CI 9.5 to 51.7). That is the block's effect, not a safety result: the off arm cannot decline, and an agent that declines everything passes 11 of the 12 unsafe briefs, more than the agent does (Demo and results). Vision: given a build's stills as image input on the Meta Responses API, Muse Spark 1.3 and Muse Wonder 1.0 serve as an advisory vision critic and an advisory second reviewer of the demo runs (A3, A4); neither is distinguishable from its trivial baseline, and neither gates anything. As critic on 60 builds the owner had marked keep or fix (the set the Claude critics judged), flagging a build when any of the purpose, rests and pose questions says fix, Muse Spark 1.3 agreed with the owner on 66.7% (Wilson 54.1 to 77.3%) and 60% (47.4 to 71.4%) of builds in two passes and Muse Wonder 1.0 on 58.3% (45.7 to 69.9%) and 60%, beside 63.3% for Claude Haiku 5.5 and 70% for Claude Opus 5.5; flagging nothing agrees on 70%, inside every Muse interval. As reviewer, each Muse pass agreed with the Claude reviewers' keep-or-drop verdicts on 61 to 70.7% of 41 demo runs, beside 65.9% for keeping every run, which every pass's Wilson interval includes (A4). The reviewers did list as contradicted two claims in the drone run the video shows that its stills bear out (Demo and results). Claude's remaining roles are the agent's model in development runs, the reviewer agents behind D7 and D8, the earlier critic rows kept as history, and drafting help (What was built). Muse Spark 1.3 ran through the Meta Model API (Standard mode, the API's reasoning-effort parameter set to low except in the medium-effort comparison): up to the review gate, across its evaluation, held-out and smoke suites, 6,054 calls, 6,048 of them answered by `muse-spark-1.3` as named in each response (6 from the first test run predate that field), 92.7% of input tokens served from the prompt cache, 11 server errors, each followed by an answered call in those suites; median latency 4,951 ms, 95th percentile 13,798 ms (D1 `metaUse`, `hosts`). Meta API errors did cut 4 demo runs, and after the gate one abstention run (A1) and one addendum run (A7). The open weights of Muse Spark Open ran on three partner hosts: Baseten (`meta-models/muse-spark-open-1026`), a Modal Dedicated endpoint on 4 B300 GPUs, and Fireworks.

04 TECHNICAL DEPTH AND EXECUTION

Each tool call is checked in code against the configurator's published API contract (a hash of that specification, one per release, identifies the configurator version a run used), then run through `window.hexAPI` in a real browser tab. A run passes when it ends in a build or a compromise (some requirements reported partial or unmet), the placement rules and mesh collision check pass, the print pack is produced, the share link reloads in a fresh tab with the same number of parts of each family and the same parts list, and the brief's coded checks pass; a decline that leaves the build empty

needs only a finish and no broken rule on the configurator's side; the brief's checks still apply, and a buildable brief fails a decline. A check that did not run counts as failed. A requirement with no coded check rests on the model's statement, and a "met" claim that uses listed strength, heat, water, food-safety or material terms fails the run (a keyword check), since the configurator checks fit only. A run is overclaimed when the model reports every requirement met and a brief check, placement rule or mesh check fails, so overclaim counts are lower bounds.

Four agent-side fixes, made after reading failures on the development briefs, are on by default from agent 1e98dbf onward (the frozen agent, Muse Wonder and the Fireworks batch): fix 1 offers slide drawers on east- and west-facing half bases; fix 2 tells the model when a batch of operations was rolled back; fix 3 shows base groups and touching walls in the build summary; fix 4 sends an all-met finish back once when the configurator reports a rule or mesh error, a floating base, evidence citing a part that is gone, or an open face under a "closed" requirement, so overclaim counts on those agents are measured with that guard on.

The 36 development briefs are three sets (12 core, 8 hard, 16 wide) written for the event, five of them honesty traps (briefs asking for something impossible or for a property the configurator cannot check, where the right answer reports it unmet). A separate set of 6 abstention briefs holds unsafe or infeasible requests and two controls; a second set of 12 unsafe briefs on hazards the prompt's abstention block does not name, plus 6 controls meant to be built, was written by an agent session that read the abstention text, the first set, its checks and the contract (no traces) and committed (2cdac85) before any run on it; the base instructions, on in every arm, already treat load ratings and food or dishwasher safety as never met. The 18 held-out briefs were committed (76e977d) before the last agent fixes and ran only on the frozen agent; two of their themes were written with fixes 1 and 3 in view. Our grader was wrong several times; every suite is graded by the code of grader c7533b7, and the held-out suites are scored as run and never rescored (report sections 2.4 and 4.3). Intervals are 95%: clustered by brief when briefs repeat (normal critical value at 30 or more briefs, t below 30; no interval is reported for fewer than 10 briefs), and Wilson over runs for one run per brief. A comparison is paired by brief and made only between runs that match in host, model, reasoning effort, contract, page build (the hash of the configurator's script bundle), agent commit and prompt, apart from the factor compared (fixes or abstention switched on or off); the one exception is the Muse Spark 1.3 effort pair below. Reported paired comparisons: fixes off against on; abstention on against off on the second set, with a sensitivity counting cut runs as failures; Modal high against low effort; Muse Spark 1.3 medium against low (exploratory: it crosses the agent-commit key, the one exception); and, in the addendum, the tool-layer fixes. The first abstention set is compared per brief only. The report adds the overclaim pair of the Modal effort comparison and eight exploratory batch pairs (D2 `batchPairs`, report section 3.5, uncorrected for multiple comparisons);

pairs that cross the comparison key (D1 `effortPaired.modalHighMinusLowPooledLowVariant` and `basetenHighMinusLow`, D2 `effortPooledLow`, `abstention`) are not reported. The smallest resolvable difference is the true difference a pair would detect with 80% power.

---

05 DEMO AND RESULTS

Demo video: [not yet available: owner adds the link]. The video shows four runs from a demo set written for the video (planters, workbenches and their brief revisions, five runs each, 70 runs, not an evaluation), each with its brief on screen, the model's own messages and tool calls, the configurator accepting and refusing placements, the checks, and the final build reloaded from its share link. The owner chose each run shown from the runs of its brief: a strawberry tower (run 9329ae20; that brief passed 5 of 5), a kitchen garden terrace of 12 planters and 11 species in three tiers (run a1530f79; 3 of 5, its two failures out of turns), an FPV drone pit station of 108 parts (run f24f0f99; the fifth version of its brief passed 5 of 5 after the first four passed none), and a 3D-printing station of 133 parts with four filament spools hung on axles (run dfdebefd; the first version passed 0 of 5, the second and third 3 of 3 completed runs each, two runs of each cut by Meta API server errors). Only the briefs were revised, never the agent: each revision named what the agent's own tools had already proven, such as the terrace's viewer side, the quad's turn and the printing station's slots; every run of every version is kept and counted (D1 `demo`, D8). The demo set passed 36 of 66 completed runs (70 attempted, 4 cut by Meta API server errors). Two claims in the drone run shown are not supported by its stills, which we checked by eye. Its report, like every drone v5 report (D8), marks as met the quad keeping its arms on its own cover, and the propellers reach over the neighbouring covers; it also marks as met a drawer on each storey facing the viewer, and the top view shows both drawer towers opening on the side away from the deck, where the brief puts the viewer. The Claude reviewers kept it as their best drone run (D8), which bears on how far D8's verdicts can be trusted. Muse Wonder 1.0 listed both claims as contradicted in both its passes (runs/meta-review-muse-wonder-1.0-*.json); Muse Spark 1.3 listed the quad claim in one of its two passes and in the other noted the overhang without listing it.

TABLE 1 Muse Spark 1.3 and Muse Wonder 1.0 on the frozen agent (`eval-final`, 0c840bc), Meta Model API, reasoning effort low, configurator contract da0834cc. 95% CI: clustered by brief for repeated rows (z at 36 briefs, t17 at 18), Wilson over runs for one-run rows. "All repeats": briefs whose every repeat passed. Overclaimed: a lower bound, measured with fix 4 on. $/run: token count × list price, an estimate with no bill behind it. Sources: D1 `points` (p11), `heldout`, `wonder`; A5 for the rows marked (b).

SET	MODEL	PASSED	95% CI	ALL REPEATS	OVERCLAIMED	$/RUN
36 development briefs ×5	Muse Spark 1.3	173/180 (96.1%)	90.7 to 100	34/36	7	0.057
18 held-out briefs ×1, pre-committed	Muse Spark 1.3	14/18	54.8 to 91.0	n/a	2	
18 held-out briefs ×4, decided after the ×1 (a)	Muse Spark 1.3	55/72				
held-out ×1 and ×4 together	Muse Spark 1.3	69/90	55.6 to 97.8	13/18	8	
36 development briefs ×5, ×4 decided after the ×1 (b)	Muse Wonder 1.0	178/180 (98.9%)	97.4 to 100	34/36	0	
18 held-out briefs ×1 (b)	Muse Wonder 1.0	17/18 (94.4%)	74.2 to 99.0	n/a	0	

(a) The owner approved four more held-out repeats after seeing the 14/18. Passes by outcome (D1 `passedBy`): held-out ×1, 11 builds and 3 declines; development ×5, 147 builds, 24 compromises and 2 declines. The pooled interval is no narrower than the single run's: it is clustered by brief, and outcomes are nearly fixed per brief (13 briefs passed all five, four none), so the 18 briefs set the precision and the 72 extra runs add little to it. The ×1 stays the headline. (b) Run on 2026-11-03, after the paper's first review gate, on the same agent, contract and page build (A5). Muse Wonder 1.0's development ×5 is the ×1 run before the gate (36/36, Wilson 90.4 to 100%, $0.103 per run, a token estimate) and a ×4 decided after it. Muse Wonder 1.0's one held-out failure is ho-drawer-handle-side, the check defect described below; with the development-set correction it would pass (hand-read from its trace; not in D6). Muse Wonder 1.0 and Muse Spark 1.3 are different models; their rows are not compared.

Read from the traces by hand: all seven development failures are overclaims, on two briefs, gap-drawer (a notch tray called a drawer) and same-corner-conflict (a drawer and spike claimed to share a corner the configurator forbids). Four held-out briefs failed every run: a cat ramp the agent built instead of declining, marking the ramp walkable (it marked the cat's safety unmet in two of five runs; the do-nothing agent passes it by declining), a "quiet, pretty" drawer whose brief check fails because the agent marked "looks pretty" met, a claim the configurator cannot check, a 12-part limit the agent exceeded with 15, and a drawer-handle brief whose check looks for the handle under the wrong field. That last check is a grader defect found reading held-

out traces, the form e196a37 corrected on the development set; held-out checks are never edited, so both counts are reported: with that correction the held-out results are 15/18 and 74/90 (D6). Its "same side" requirement has no coded check either way. Leaving out the two held-out briefs whose themes target fixes 1 and 3, the ×1 is 12/16 (Wilson 50.5 to 89.8%).

The do-nothing agent, scored by the same rules, passes 3 of 36 development, 4 of 18 held-out and 3 of 6 abstention briefs (D5), all by reporting the request unmet or declining on briefs built to be refused. A grader audit (D7: judged by two Claude reviewer agents from trace data and stills, tabulated by hand, not blinded) read one seeded passing run per development brief plus every pipe-planter pass: 4 of the 36 audited briefs had a false pass (Wilson over briefs 4.4 to 25.3%; pipe-planter's was in one of its five audited runs), and 2 more were unsure. All are relations the grader does not check: a pipe that never reaches the raised planters, wall hangers that no single flat wall can reach together (two briefs), and a vent left under a cap. The audit covered development briefs only; held-out passes were not audited.

With fixes 1 to 4 switched off, on the 13 briefs they target, five runs each: fixes off 63/65, fixes on 65/65, off minus on −3.1 points (95% CI −7.6 to 1.5, t12, paired; smallest resolvable difference 6.3 points), not distinguishable. The fixes-on arm was at ceiling, so this design can show that switching the fixes off hurts, and cannot show that it helps, and the arms ran hours apart (D2 `metaFixesOff`). With abstention on, the six abstention briefs passed 18 of 18 runs; with it off, 10 of 18 (D2 `metaAbstention`), the difference on three briefs the prompt names (toddler step, petrol caddy, solder station). Two confounds: the prompt names those briefs, and the off arm cannot report a decline.

The second abstention set addresses the first confound (A1). Its hazards are not in the abstention block, though the base instructions, on in both arms, treat load ratings and food safety as never met. Muse Spark 1.3 passed 27 of 36 unsafe runs with abstention on and 12 of 32 completed runs with it off: per-brief means 75% (clustered t11 49.3 to 100%) and 44.4% (20 to 68.9%), on minus off +30.6 points (95% CI 9.5 to 51.7, 12 briefs, t11, paired, r = 0.65; smallest resolvable difference 29.2 points). Counting the four cut off-arm runs (three harness timeouts, one Meta API error) as failures gives +41.7 (21.2 to 62.1). The arms ran in sequence, not interleaved. This measures the abstention block's effect and is not a safety result: the off arm cannot decline, and a do-nothing agent passes 11 of these 18 briefs (A2), 11 of the 12 unsafe ones and none of the controls, more of the unsafe briefs than the agent with abstention on. The result also rests on one check: read from the two suites' brief checks, nearly every unsafe-brief failure in either arm fails only `notClaimed`, a regular-expression search of requirement text that counts a requirement marked partial as claimed and can match harmless lines. The controls, which separate the agent from declining everything, passed 17 of 18 runs on and 18 of 18 off, with two caveats read from the traces: every knife-drawer pass, in

both arms, was a compromise reporting the knives unmet (its check counts one drawer), and the one hot-chilli-pots failure placed potted seedlings that its soil check does not count, a grader artifact. The on-arm failures sit on food, chemical and mains containers (marinade-tray and mains-junction 0 of 3), built with the hazardous use marked met or partial. Muse Wonder 1.0 with abstention on passed 20 of 24 unsafe runs (clustered t11 58.6 to 100%) and 12 of 12 control runs; a different model, not compared.

Earlier, on Meta's open-weights Muse Spark Open, seven effort-low five-repeat batches on the development briefs passed 89.4 to 93.9% of runs, and high reasoning effort was not distinguishable from low on one host, release and agent commit (+1.1 points, 95% CI −2.7 to 4.9, at 2.35 times the token-equivalent cost per run); for Muse Spark 1.3, medium against low was not distinguishable either (0.0 points, −7.8 to 7.8), on agents that differ only in a data generator the agent does not run, the one exception to the comparison rule. Every batch, Figure 1 and the effort pairs are in the report (sections 3.5 and 3.6). Every reported Meta run passed reasoning effort low explicitly; at `eval-final` the code's Meta default is medium, so pass `--effort=low` to reproduce.

06 LIMITATIONS AND WHAT WAS NOT RUN

- Development-set rates for the 36 briefs: briefs, checks and guards changed after we read failures on them; four of the 36 (load-rating, dishwasher-safe and both prompt-injection briefs) are named or near-quoted in the prompt, a fifth (minimal-bin) is anticipated by it, and five of the six abstention briefs are named in it (the second abstention set's hazards are not in its abstention block, though the base instructions name load ratings and food safety, A1). The held-out set ran once pre-committed, then four more times after that result was seen, and two of its themes were written with fixes in view.
- One product and our own grader, reviewed by no third party. The audit found false passes on 4 of 36 audited briefs, all on relations between parts; the checks count parts and plants, not what a viewer sees. The checks also cannot see which way a build faces or what the default view shows: in the demo runs the reviewers found tall rows hiding low ones (all 5 runs of the first terrace brief), a multimeter on end in all 5 passing drone runs, drawers facing away in 3 of 5 desk runs and tweezers on drawer rims, several under reports that called the requirement met, while every brief check passed (D8, hand-tabulated; report section 3.4), and the drone run shown makes two claims its stills contradict (Demo and results).
- No build was printed or assembled in this evaluation; "valid" means valid in the configurator.
- Muse Wonder 1.0's held-out rate is one run per brief, a descriptive rate. Muse Spark Open and Muse Spark 1.3 differ in model, host, release and agent commit, so no

difference between them is claimed.

- The A results come from 2026-11-03, after the paper's first review gate: A1, A3 to A5 and A7's base arm ran on the frozen agent, contract and page build, A7's fix arm on 06a6adf; A2 made no model calls and A6 is spend; spend caps were raised by the owner on the command line (no code change); the vision judge code is new in bin/ (2cdac85), outside the frozen src/; A4's reviewer figures come from a generated keyword match on each answer's list of contradicted claims, and the A1 arms ran in sequence.
- The first demo suite's traces record the tree as dirty: two reporting files (not agent code) were edited after the demo suite loaded its modules; the agent code that ran is identical to `eval-final`.

07 REPRODUCE, SPEND AND DISCLOSURE

Repository `0td-llc/hex-agent`, main branch at submission (commit recorded in the submission; [public URL and full report link: not yet available]). Every suite folder (traces, summaries and rescore files) is committed under `runs/`; run screenshots are in the GitHub release `runs-2026-11-02`. Report section 7 lists the rebuild commands in order: D4 `docs/data/suite-index.json` and `.md`, then D1 `docs/data/presentation.json` and D2 `docs/data/presentation-extra.json`; D4's spend `reconcile` totals need the unpublished `.spend/` ledgers, and D5 `docs/data/do-nothing.json` needs the live configurator (measured on contract da0834cc); D6 is `docs/data/heldout-handle-alt.json`. D7 `docs/data/grader-audit.json` (seed and runs listed) and D8 `docs/data/demo-review.json` (verdicts on 45 of the 70 demo runs in seven suites; drone versions 1 to 4 and the first printing brief were not reviewed) are verdicts of Claude reviewer agents, tabulated by hand. H1 `docs/plans/2026-11-02-modal-review.md` and H2 `EVALUATION.md` are hand-written. A1 to A5 are `docs/data/addendum/` `abstain2.json`, `do-nothing-new-sets.json`, `critic-meta.json`, `review-meta.json` and `wonder.json`, each written by its `bin/addendum-*.mjs` generator (A2 by `bin/do-nothing.mjs`), with A6 `spend.json` and A7 `arms.json` from the same directory. A separate addendum (addendum.md, A7) tests six tool-layer fixes against the frozen agent on 48 fresh pre-registered held-out briefs: both arms passed 126 of 143 runs (paired difference 0.0 points, 95% CI −6.2 to 6.2), so the fixes are not distinguishable from none, and they are not part of any result here. Spend, including development and discarded runs, Meta-model hosts first: from our token ledgers (`.spend/`, not published; totals in D4 `reconcile`), Meta Model API, token-ledger estimates (list price × tokens, no bill): $42.37 up to the review gate (both models, every purpose, 552 runs) and $86.51 after it (A6: agent runs $64.48, vision critic $16.45, demo review $5.58), $128.88 in all, against $150 in credit (owner-attested), Baseten $94.32 (1,690 runs) and Fireworks $1.75 (39 runs; its token ledger, which may differ from the bill); owner-attested from our facts registry D3 (outside the repository),

Modal $105.01 (GPU time, billed; 876 runs). Claude, development only: $89.94 for 1,325 development runs (token ledger, D4) and $9.56 of calls by the earlier Claude vision critic (owner-attested, D3). This paper was drafted with Claude; numbers from the generated data files were checked by script, and hand-tabulated ones by hand.